Cameras for autonomous driving

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Outline

- Introduction
- Camera parameters
- Exercises

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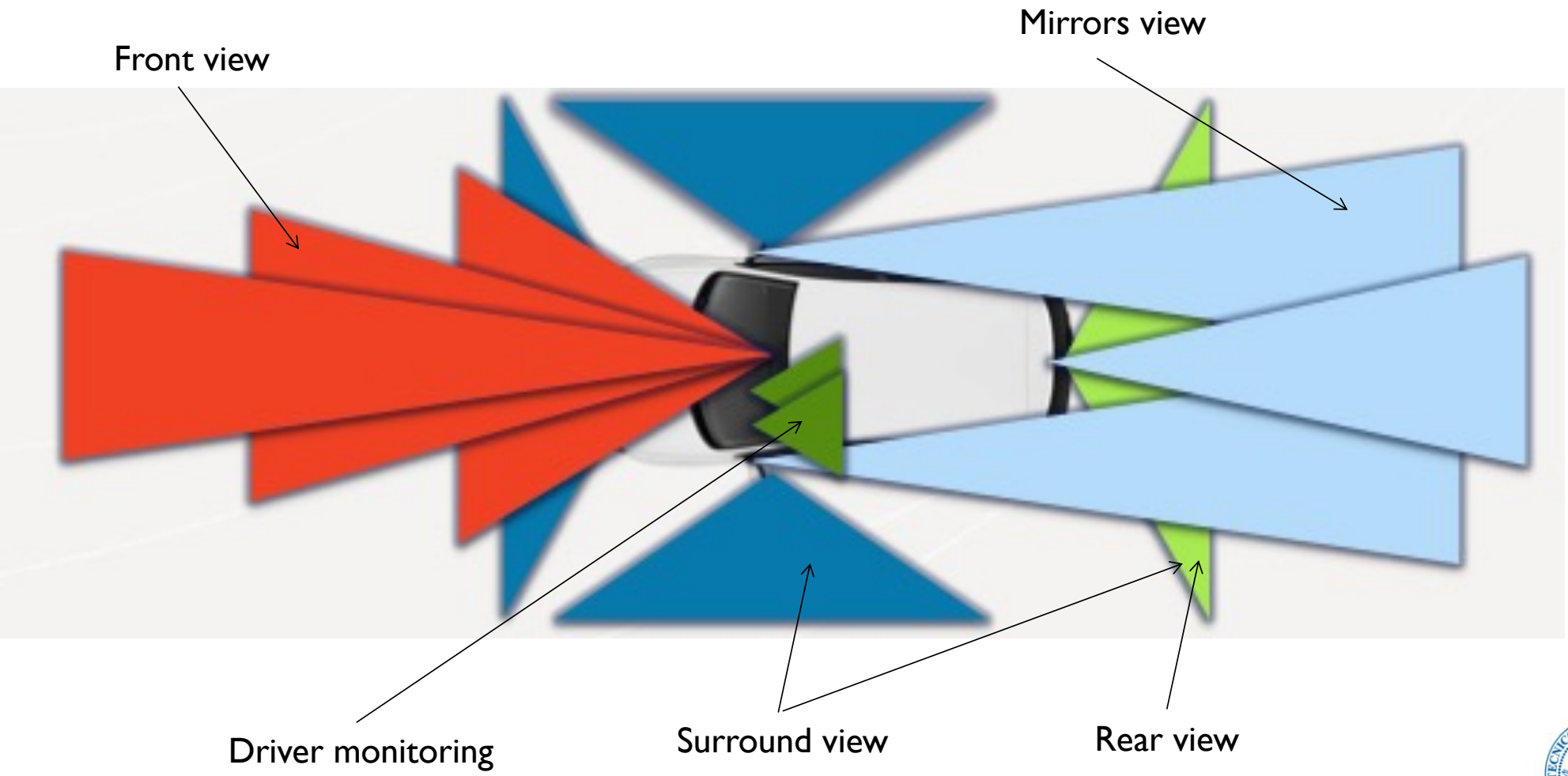
Introduction

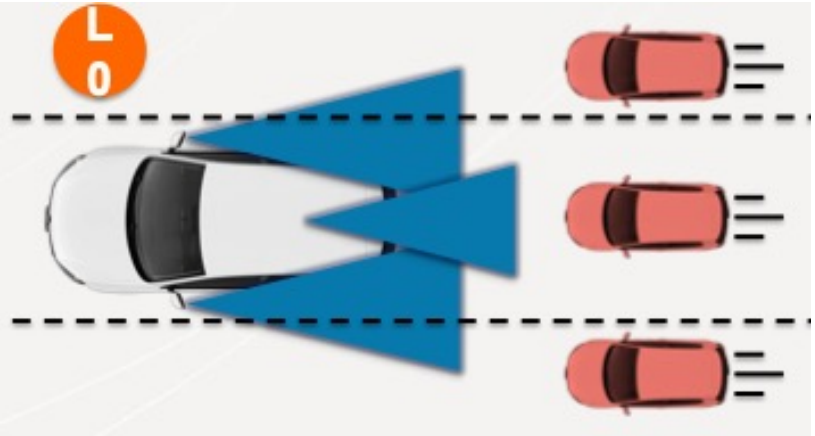
- Cameras serves multiple purposes
- Enhance visibility
- Driver monitoring
- Provides inputs for object and event detection and response systems

Camera locations



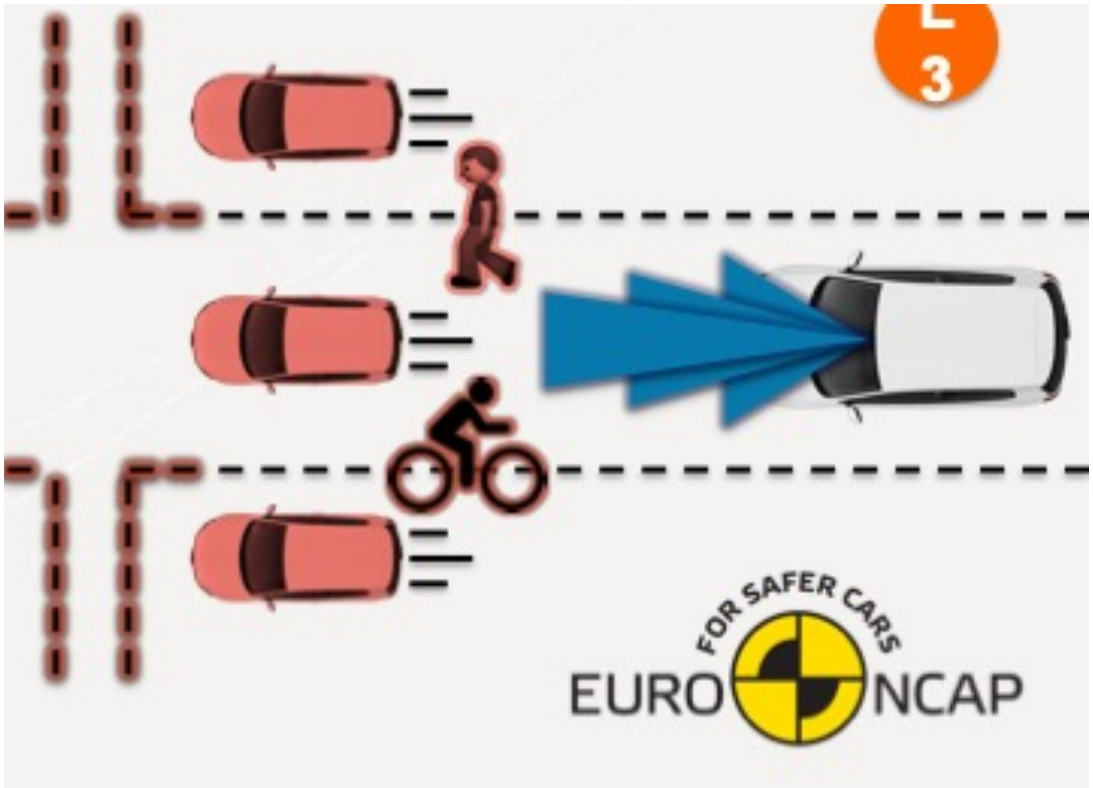
Enhance visibility – possibile usage

- Parking assist
- Cross traffic alert
- Backover protection
- Trailer steering assist
- Blind spot detect
- Rear collision warning



Enhance visibility – possibile usage

- Emergency braking, auto cruise control
- Pedestrian and vehicle detection
- Lane detection and keeping
- Traffic sign recognition
- Headlight control
- Bicycle recognition
- Intersections



Driver monitoring

- Driver drowsiness detection
- Driver distraction detection
- Airbag deployment
- Seat belt adjustment
- Driver authentication

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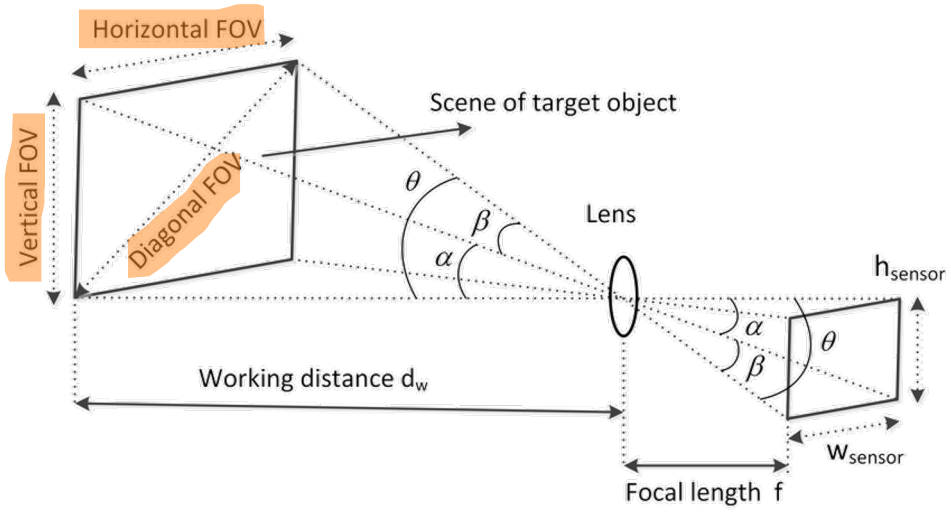
Camera parameters

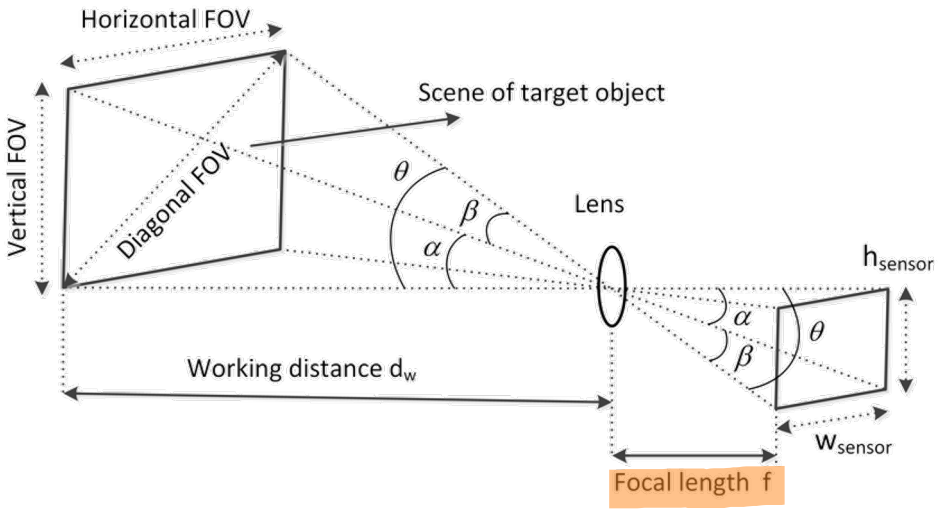
- Detection range, it is the distance that a camera can detect an object
- Angular resolution, it is the ability to distinguish details of an object
- Field of view, it is the extent of the observable world that is seen at any given moment



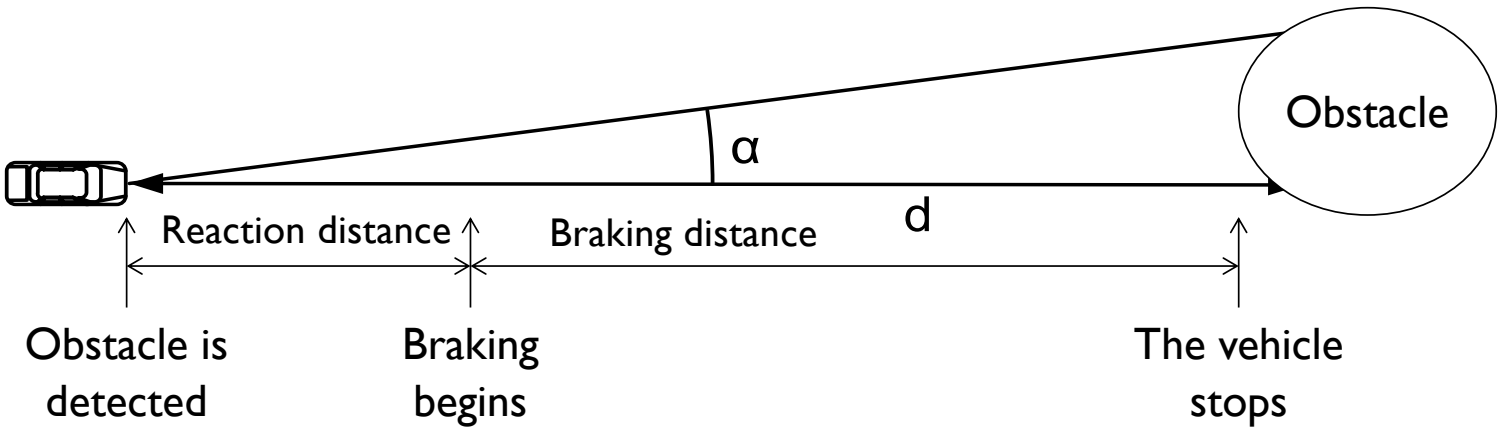
Camera parameters

- Frame rate is the number of consecutive images that can be taken in one second (from 24 to 60 fps)

- Focal length, distance between the lens and the image sensor when the subject is in focus



Detection range



$$d \geq Reaction\ Distance + Braking\ Distance$$

$$Reaction\ Distance[m] = Speed\ \left[\frac{km}{h}\right] \cdot Reaction\ Time\ [s] \cdot \frac{1}{3.6}$$

$$Braking\ Distance\ [m] = \frac{Speed^2[\frac{Km}{h}]}{250 \cdot f} \qquad f = \begin{cases} 0.8\ for\ dry\ asphalt \\ 0.5\ for\ wet\ asphalt \\ 0.3\ for\ snow \\ 0.2\ for\ ice \end{cases}$$

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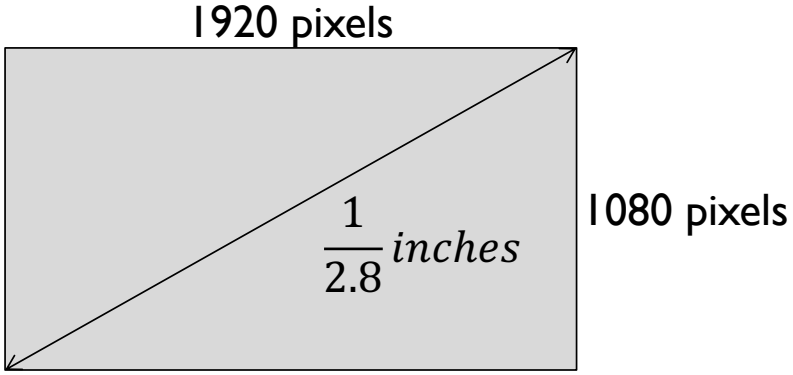
Exercise

- A sensor with the following characteristics is selected for an emergency braking system

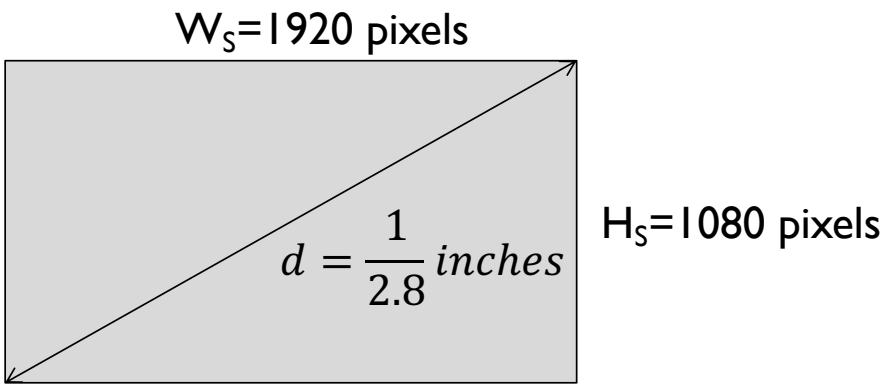


- The system identifies 1m high x 0.5m wide obstacles in the city (50 Km/h speed): 40,28m

- You shall:
  - Select a suitable focal length
  - Compute the number of pixels a 1m x 0.5 m object at 40m takes on the sensor

Solution

- We need to compute the geometry of the sensor
  - Width: W_s
  - Height: H_s
  - Pixel dimension (we assume a square pixel): p



$$d = \frac{1}{2.8} inches = \sqrt{{W_S}^2 + {H_S}^2}$$

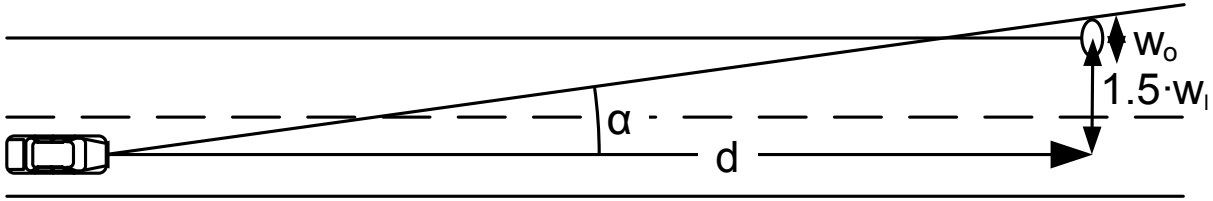
$$W_S = \frac{1920}{1080} H_S$$

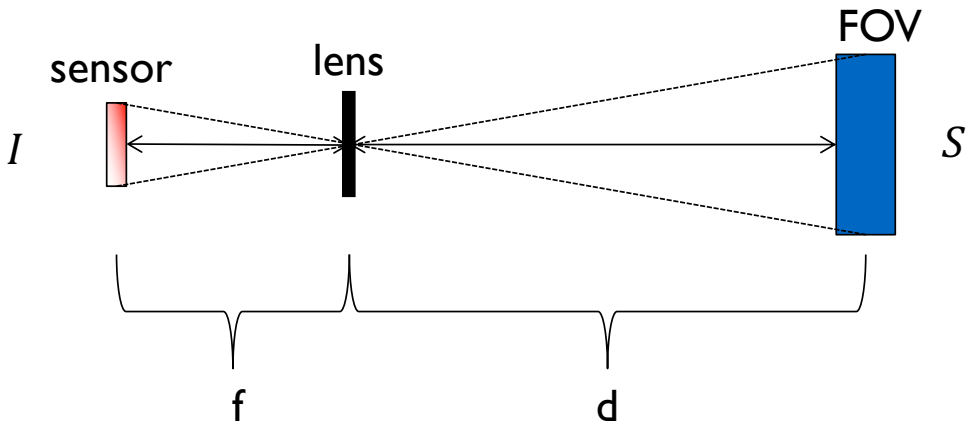
$$W_S = 7.91mm$$
$$H_S = 4.44mm$$

$$p = \frac{7.91}{1920} = 0{,}0041mm$$

$$p = \frac{4.44}{1080} = 0{,}0041mm$$

Solution

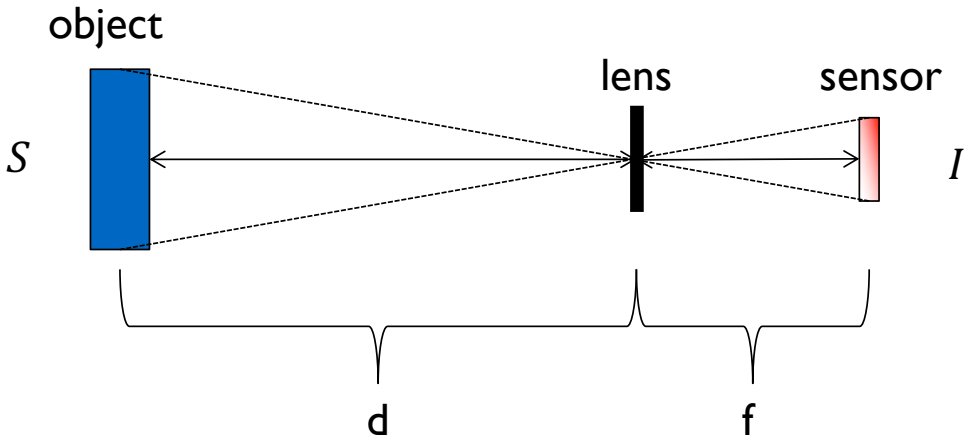




$$\frac{S}{d} = \frac{I}{f} \qquad \frac{1{,}5x3{,}75 + 0{,}5/2}{40{,}28} = \frac{1920x0{,}0041}{f}$$

$$f = 53{,}97mm \rightarrow f = 54mm$$

Solution



$$\frac{0{,}5m}{40{,}28m} = \frac{I}{54mm} \Rightarrow I = 0.67\ mm$$

$$I = \frac{0.67mm}{0.0041mm} = 163pixels$$

Obstacle 1m x 0.5m @ 40,28m = 326x163 pixels

$$\frac{S}{d} = \frac{I}{f}$$

Numerical exercise

- The following camera set-up is being considered for an Emergency Braking system:
  - 8 megapixels Sony IMX219 sensor: 3280 x 2464 pixel;
  - 1.55µm × 1.55µm pixel size;
  - 8 mm lens.
- Let's assume the smallest object that shall be recognized is 0.75m wide and 1.5m tall, and that it shall produce an image that is at least 20x20 pixels to be recognized.
- Assuming a system reaction time of 1 s, which is the maximum speed a vehicle could travel for the Emergency Braking system to avoid the accident (i.e., bringing the car to a full stop after the obstacle is recognized)?